

A4 · Answer Key and Demo Script

CS425 · Where Does This Datagram Go? · instructor led

INSTRUCTOR COPY

Every transcript below is real output captured on `onyx.boisestate.edu` on October 4, 2026. Hops and times will drift; the shapes will not. About 40 minutes: 10, 10, 8, 8, then 4 for the exit question.

Say this at the start

"Every router on the Internet answers one question for every packet: which way does this go? Onyx is not a router, but it answers the same question the same way, so today we ask it. Everyone `ssh onyx`. Predict before we run anything." Then hand out the sheet.

ROUND 1 · ONYX'S ADDRESS

ip -4 -br addr		
lo	UNKNOWN	127.0.0.1/8
eno12399np0	UP	132.178.227.11/25

SAY "A /25 is 25 network bits and 7 host bits. Do not recall it, derive it."		

- Subnet size. $2^7 = 128$ addresses, 126 usable (network and broadcast are reserved).
- Binary. $11 = 0|0001011$. The mask's last octet is $1|0000000 = 128$, so the mask is 255.255.255.128.
- Range. Network 132.178.227.0, broadcast 132.178.227.127, usable .1 to .126, 126 hosts.
- .200? No: $200 = 1|1001000$, first host bit is 1, so it is in 132.178.227.128/25. Private? No. RFC 1918 is 10/8, 172.16/12 and 192.168/16. Onyx is public, with no NAT between it and the Internet. Worth contrasting with their laptops at home.

ROUND 2 · LONGEST PREFIX MATCH

ip route then ip route get for each destination	
default via 132.178.227.1 dev eno12399np0 proto static metric 100 132.178.227.0/25 dev eno12399np0 proto kernel scope link src 132.178.227.11 metric 100	
132.178.227.50 dev eno12399np0 src 132.178.227.11 uid 26544 132.178.227.200 via 132.178.227.1 dev eno12399np0 src 132.178.227.11 uid 26544 8.8.8.8 via 132.178.227.1 dev eno12399np0 src 132.178.227.11 uid 26544 local 127.0.0.1 dev lo src 127.0.0.1 uid 26544	

DESTINATION	MATCHING LINE	DIRECT OR GATEWAY
132.178.227.50	132.178.227.0/25 (also matches default)	Direct, no via
132.178.227.200	default, outside the /25 (round 1)	Gateway .1
8.8.8.8	default	Gateway .1
127.0.0.1	none in ip route: the local table	Loopback, never leaves Onyx

- Q1. Longest prefix wins: /25 beats /0, so .50 is delivered on the link. Loopback and Onyx's own addresses live in the **local** routing table, which the kernel checks first (`ip route show table local` lists 127.0.0.0/8). Point at P3: this lookup is the project.

GRADING

Pass/fail, 20 points. Pass when every box has something in it and the round 1 arithmetic was attempted in binary. Wrong predictions are fine; that is what they are for.

ROUND 3 · HOW BIG CAN A DATAGRAM BE?

```
ip link show eno12399np0 (look for mtu 1500) then two pings
```

```
PING 132.178.227.1 (132.178.227.1) 1472(1500) bytes of data.  
1480 bytes from 132.178.227.1: icmp_seq=1 ttl=254 time=0.654 ms
```

```
PING 132.178.227.1 (132.178.227.1) 1473(1501) bytes of data.  
ping: local error: message too long, mtu=1500
```

POINT AT 1472(1500): ping itself prints the payload and the full datagram size.

Biggest payload. 1,472 bytes. $1,472 + 8$ (ICMP header) + 20 (IPv4 header, no options) = 1,500.

Q2. The 28 bytes are the ICMP and IPv4 headers. The error came from **Onyx's own kernel**: with don't fragment set and a 1,501 byte datagram, it refused to send, so nothing left the machine. Without `-M` do, Onyx (or any router with a smaller MTU on the way) fragments, and only the **destination host** reassembles, using the identifier, flags and offset fields. IPv6 routers never fragment.

FAST GROUP Why `ttl=254` and not 255 from a gateway on the same link? Not confirmed; a virtual gateway address answered by a router behind it is the likely story. Discussion only.

ROUND 4 · COUNTING ROUTERS WITH TTL

```
tracpath -n mirrors.xmission.com
```

```
1?: [LOCALHOST] pmtu 1500  
1: 132.178.227.1 0.651ms asymm 2  
1: 132.178.227.1 0.357ms asymm 2  
2: no reply  
3: 132.178.1.147 1.238ms  
4: 132.178.1.98 1.301ms  
5: 209.249.232.13 1.911ms  
6: no reply  
7: no reply  
8: no reply  
9: no reply  
10: 4.69.220.109 28.173ms asymm 9  
11: 4.35.168.186 50.639ms asymm 16  
12: 166.70.1.2 50.208ms asymm 18  
13: 198.60.22.13 50.360ms reached  
Resume: pmtu 1500 hops 13 back 19
```

SAY "Hop 13 is the server, so that is 12 routers. Five of them never answered, and the packets went through them anyway."

Count. 13 hops, so 12 routers before the server. 5 hops said no reply (2, 6, 7, 8, 9).

Q3. `tracpath` sent a datagram with TTL 3. The first two routers each decremented it, the third decremented it to zero, dropped it, and sent back an ICMP time exceeded message whose source address is hop 3. The no reply routers forwarded traffic fine (later hops answered) but do not send, or rate limit, or filter, those ICMP messages. Forwarding and answering ICMP are separate jobs. `pmtu 1500` is round 3's MTU, discovered along the path.

Optional. `asymm` means the reply came back over a different number of hops than the probe went out: the return path is not the same path.

EXIT QUESTION

Any field outside the destination address, with an action. Examples: a firewall matching destination port 23 and dropping it; NAT matching the source address and port and rewriting them; a load balancer matching destination port 443 and picking a server; a rate limiter matching the protocol field (ICMP). Generalized forwarding is exactly match plus action over any header fields (sections 4.4 and 4.5).